

[SynopStudio] MS1

Video editing and summarization with artificial intelligence.

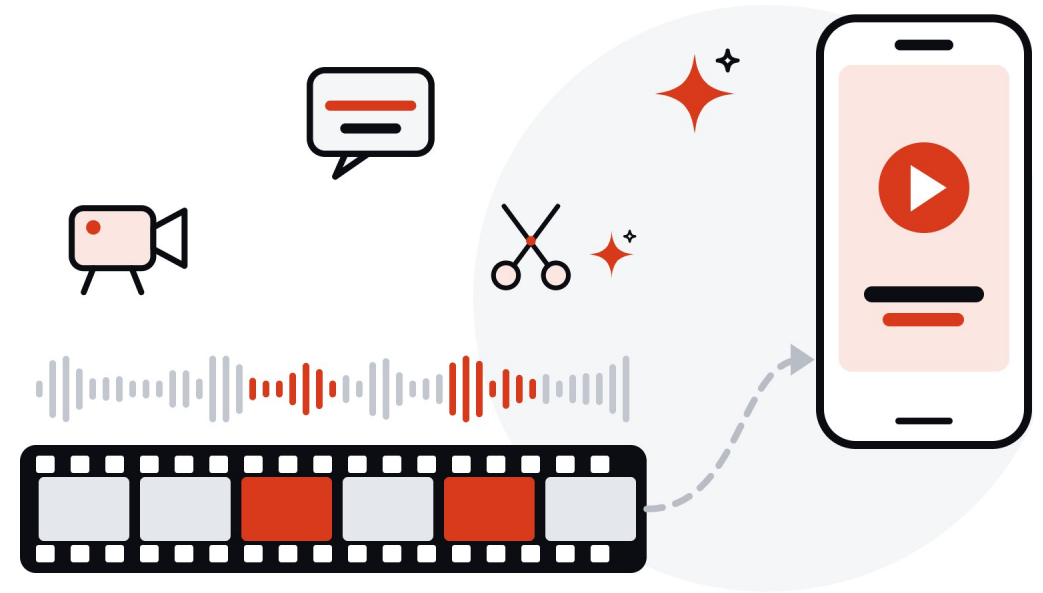
PEI – PROJETO EM ENGENHARIA INFORMÁTICA
· [2026/2027]

GROUP [15]

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[Murilo Frezzato]	[125487]
[Tomas Lopes]	[125596]

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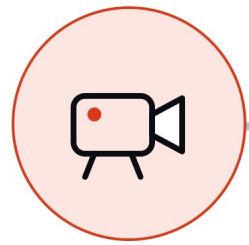


deti

universidade de aveiro
departamento de eletrónica,
telecomunicações e informática

CONTENTS

Five chapters



01

**Context and
problem**



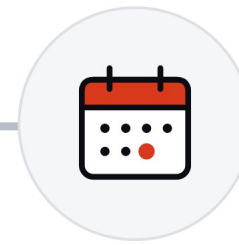
02

**Goals and
expected results**



03

Related Work



04

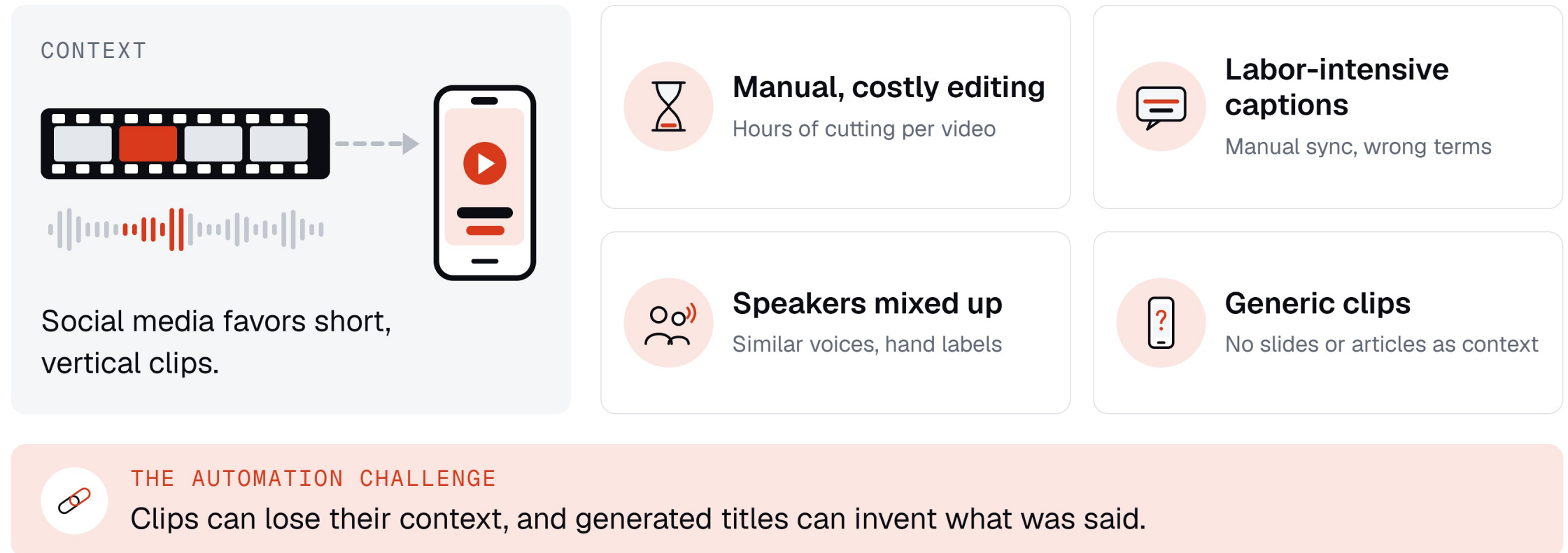
Tasks and calendar



05

**Communication
plan**

From long video to short clip

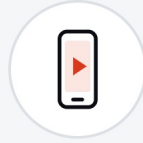


Goals and expected results



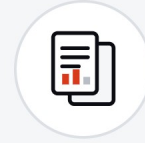
RESULTS

Expected results



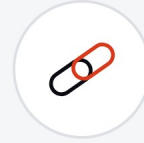
01

Trained on existing clips



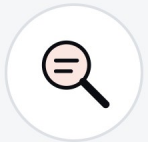
02

Context files



03

Traceable transcription



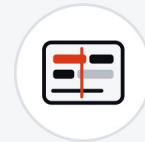
04

Topic segmentation



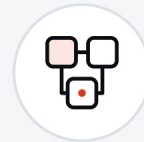
05

AI clip proposals



06

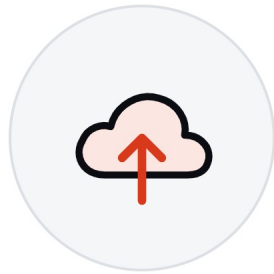
Editing from transcription



07

Modular architecture

Four steps from upload to export



01

Import

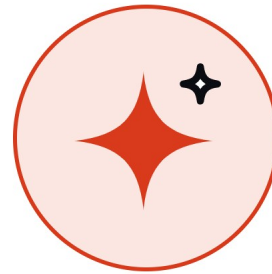
File or link



02

Transcribe

Speech to text with
timestamps



03

Summarize

The AI picks the key moments



04

Export

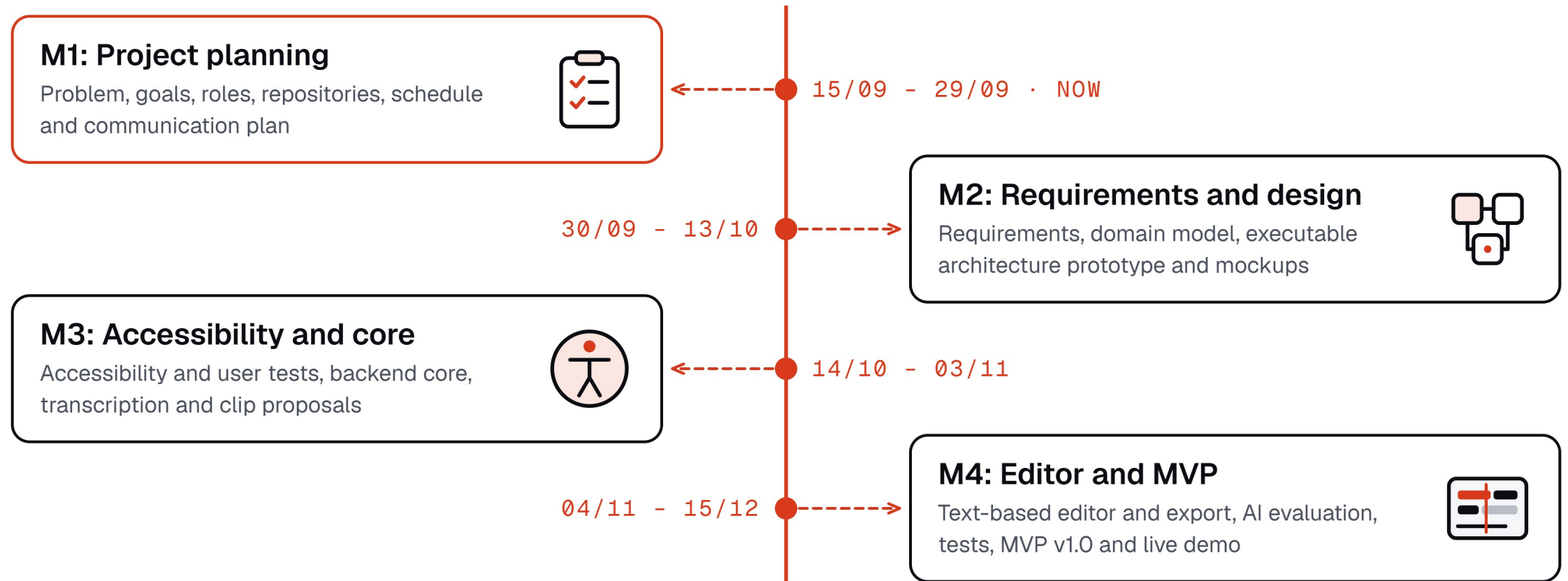
Cut, summary and captions

Related Work

✓ Supported ~ Partial/Limited ✗ Not supported

	OpusClip	CapCut	Canva
Long Video Summarization (Auto-Clipping)	✓	✓	~
Professional Timeline and Multi-Track Editing	✗	✓	~
Automated Subtitling	✓	✓	✓
Dynamic Subtitles and Keyword Highlighting	✓	✓	✗
Import of Auxiliary Contextual Documents (Slides, Articles, Reports)	✗	✗	✗
Verifiable Source Traceability and Internal Linkage	✗	✗	✗
Speaker Tracking and Vertical Reframing	✓	✓	✗

Tasks per milestone



Milestone 1

- T1.1 - Context and problem clarification
- T1.2 - Organization and repository setup
- T1.3 - Assigning roles
- T1.4 - Related work (Draft)
- T1.5 - Define Goals of the Project
- T1.6 - Project Schedule and Task List
- T1.7 - Communication Plan
- T1.8 - MS1 - Presentation

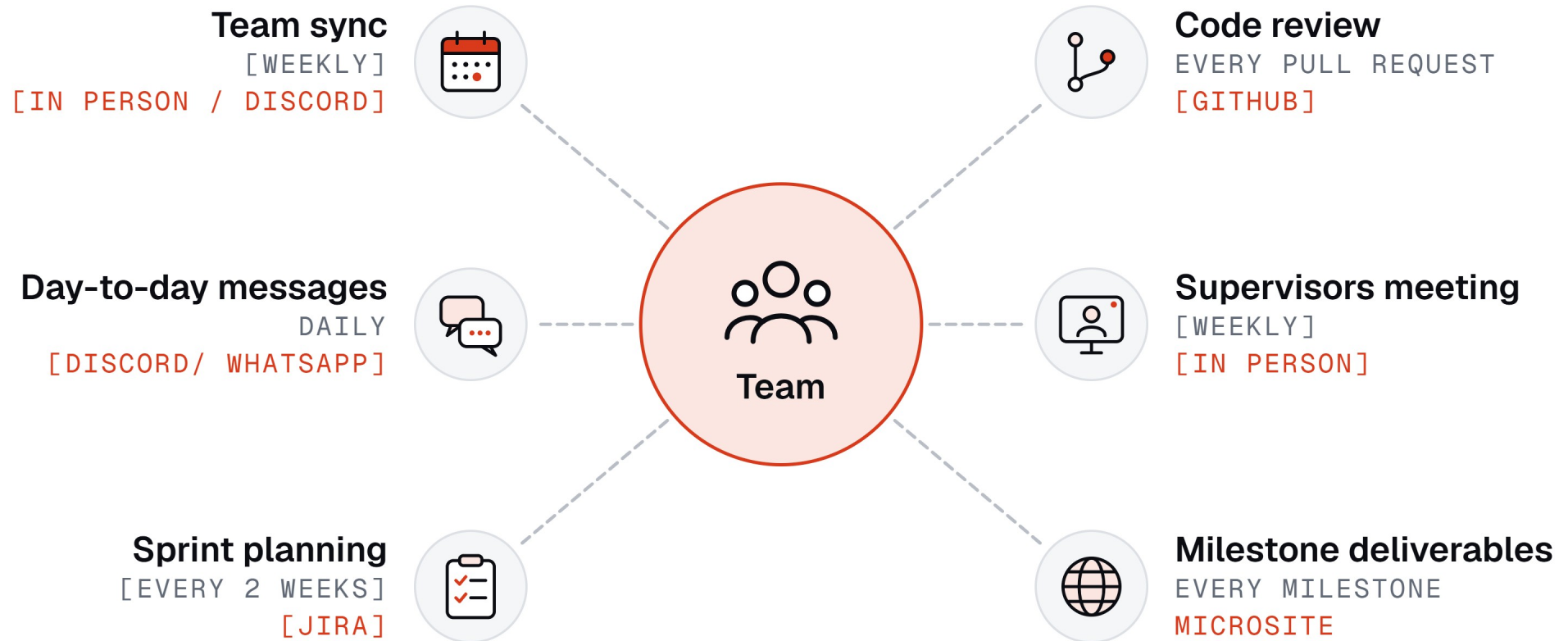
Milestone 2

- T2.1 - Gather Functional Requirements
- T2.2 - Improve State of the Art
- T2.3 - Define personas and use cases
- T2.4 - Gather Non-functional Requirements
- T2.5 - Create Domain Model and Data Contract
- T2.6 - Model and Hardware Selection
- T2.7 - Deployment Diagram (Architecture and Tech)
- T2.8 - Project Setup (Docker, CI, Database)
- T2.9 - Executable Architecture Prototype
- T2.10 - Architecture Validation
- T2.11 - Design the Mockups (UI)
- T2.12 - MS2 - Presentation

Milestones 3 and 4

	October	November				December		
	W6-7	W7-8	W8-9	W9-10	W10-11	W11-12	W12-13	W13-14
Milestone 3								
T3.1 - Accessibility Requirements								
T3.2 - Accessibility of Generated Captions								
T3.3 - Usability Tests with Users								
T3.4 - Accessibility Audit								
T3.5 - Results Analysis and Improvements								
T3.6 - Authentication and User Accounts								
T3.7 - Video Upload and Project Management								
T3.8 - Async Job Processing and Status								
T3.9 - Transcription & Speaker Identification								
T3.10 - Clip Proposal Generation								
T3.11 - MS3 Presentation								
Milestone 4								
T4.1 - Clip Proposals Page and Traceability								
T4.2 - Text-based Editor								
T4.3 - Rendering, Captions and Export								
T4.4 - AI Metadata Generation								
T4.5 - Privacy, Audit Log and Data Deletion								
T4.6 - Accessibility and Usability Fixes								
T4.7 - Evaluation Dataset								
T4.8 - AI Model Evaluation								
T4.9 - Performance and Security Tests								
T4.10 - Stabilization and Bug Fixing								
T4.11 - MVP Scope Freeze								
T4.12 - Checkpoint Demo								
T4.13 - Documentation and Final Report								
T4.14 - Demo Preparation								
T4.15 - MS4 Presentation								

How the team communicates

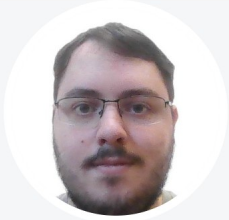


Thank you.■

Questions and feedback are welcome.

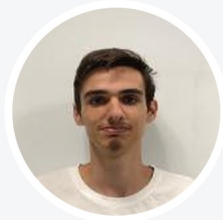


[HTTPS://PEI-SYNOPSTUDIO.GITHUB.IO/DOCS/](https://pei-synopstudio.github.io/docs/)



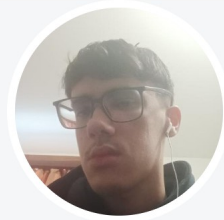
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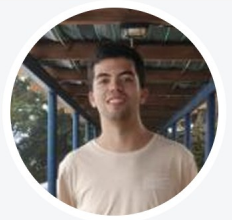
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